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Face It: how Azure Face Service can help you recognize and identify faces



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Platinum :



Great Sponsor Support

Co-Organizer : **BRAIN++**
AI FACTORY

Bronze :



AI on Azure — 10+ years of evolution



Face Service
General Availability

2009
Bing + ML
features

2017
Azure ML
goes GA

2019
OpenAI
\$1B deal

2023
AI Studio
GPT-4 + Phi

2025
Microsoft
Foundry
★ Today

2015
Cognitive
Services

2018
Bot Framework
Cognitive GA

2021
Azure OpenAI
early access

Nov 2024
Azure AI
Foundry

Pre-GenAI · 2009–2018

OpenAI era · 2019–2023

Foundry · 2024–2025

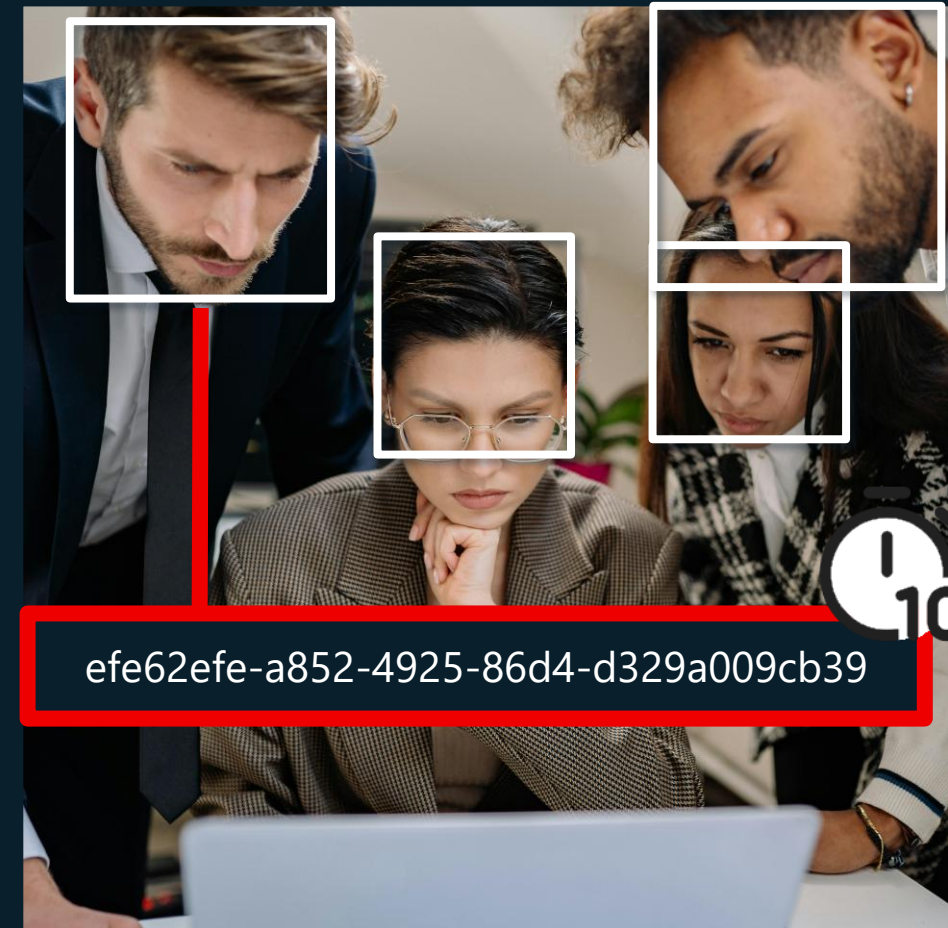
What is the Azure Face service?

- The **Azure Face Service** provides **AI algorithms** that **detect**, **recognize**, and **analyze** human faces in images.
- You can use the Face Service through a client library **SDK** or by calling the **REST API** directly.



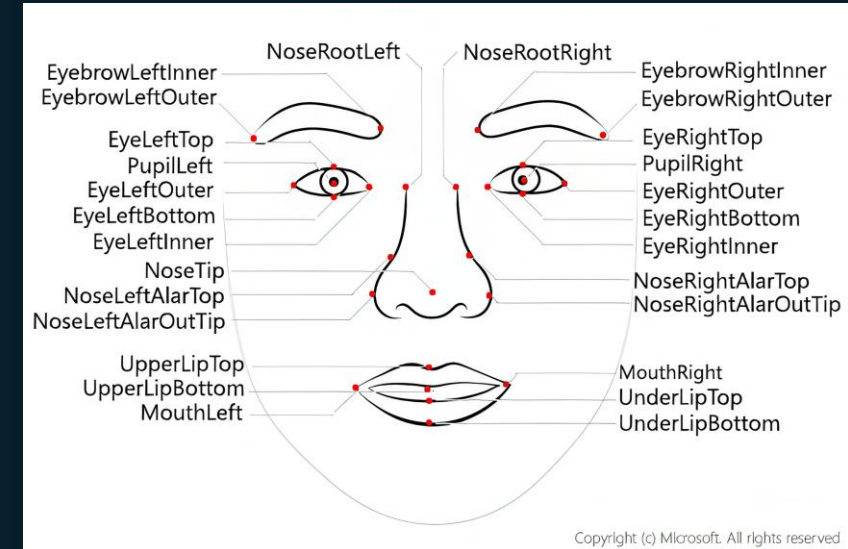
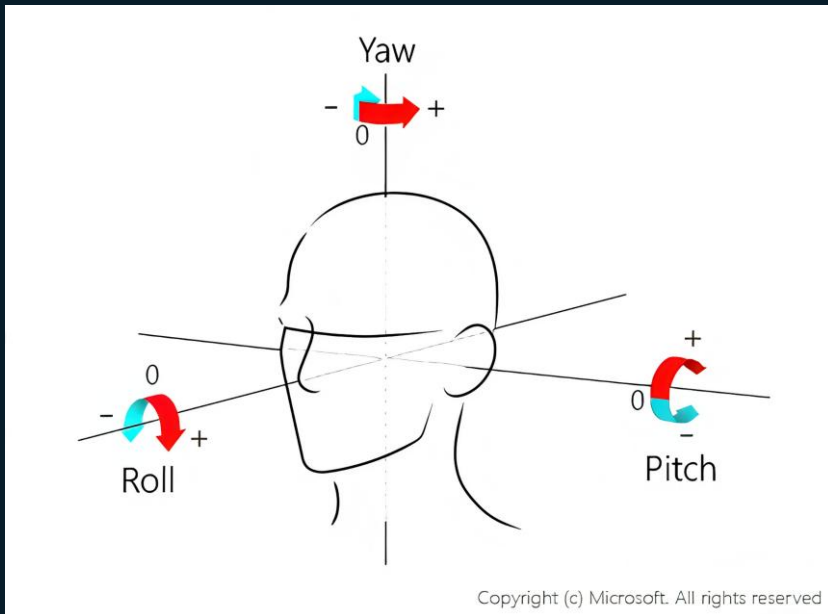
Face Detection and Analysis

- The **Detect API** detects **human faces** in an image and returns the **rectangle coordinates** of their locations.
- It also returns a **Face ID** that represents the stored face data (called **temporary faces**).
- The **Face ID** will be invalidated (and the image removed) after a period.
- It can extract a set of **face-related attributes**, such as head pose and others.



Face detection, attributes, and input data

- **Face landmarks** are a set of easy-to-find points on a face, such as the pupils or the tip of the nose.



- **Attributes** are a set of features that can optionally be detected by the Detect API.
- The supported image formats are **JPEG**, **PNG**, **GIF** (the first frame), **BMP**, image file size should be no larger than **6 MB**

Face recognition

Face recognition is the process of verifying or identifying individuals by their faces.

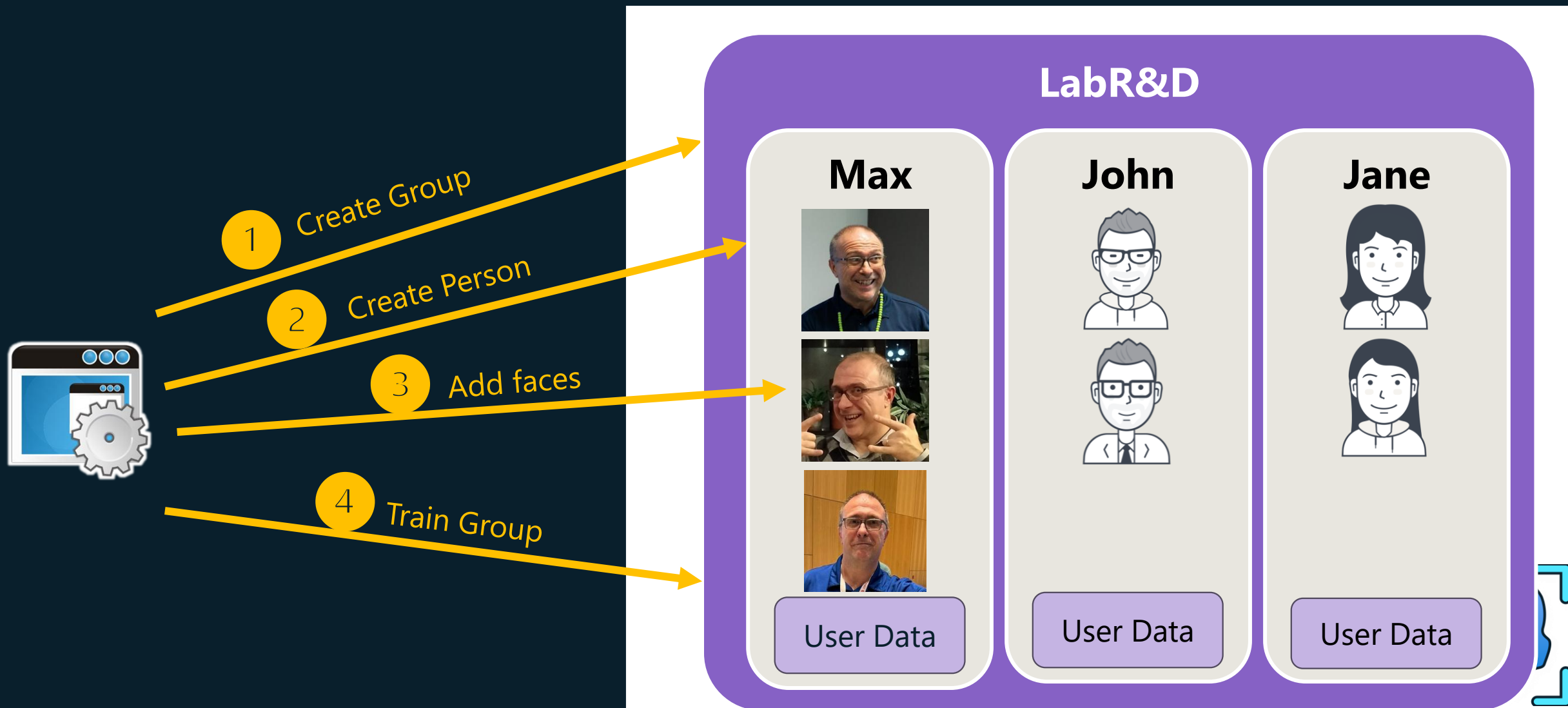
Identification

- Takes one or several source face IDs and a PersonGroup or LargePersonGroup or PersonDirectory.
- Returns a list of candidates (with prediction confidence value) that each source face might belong to.

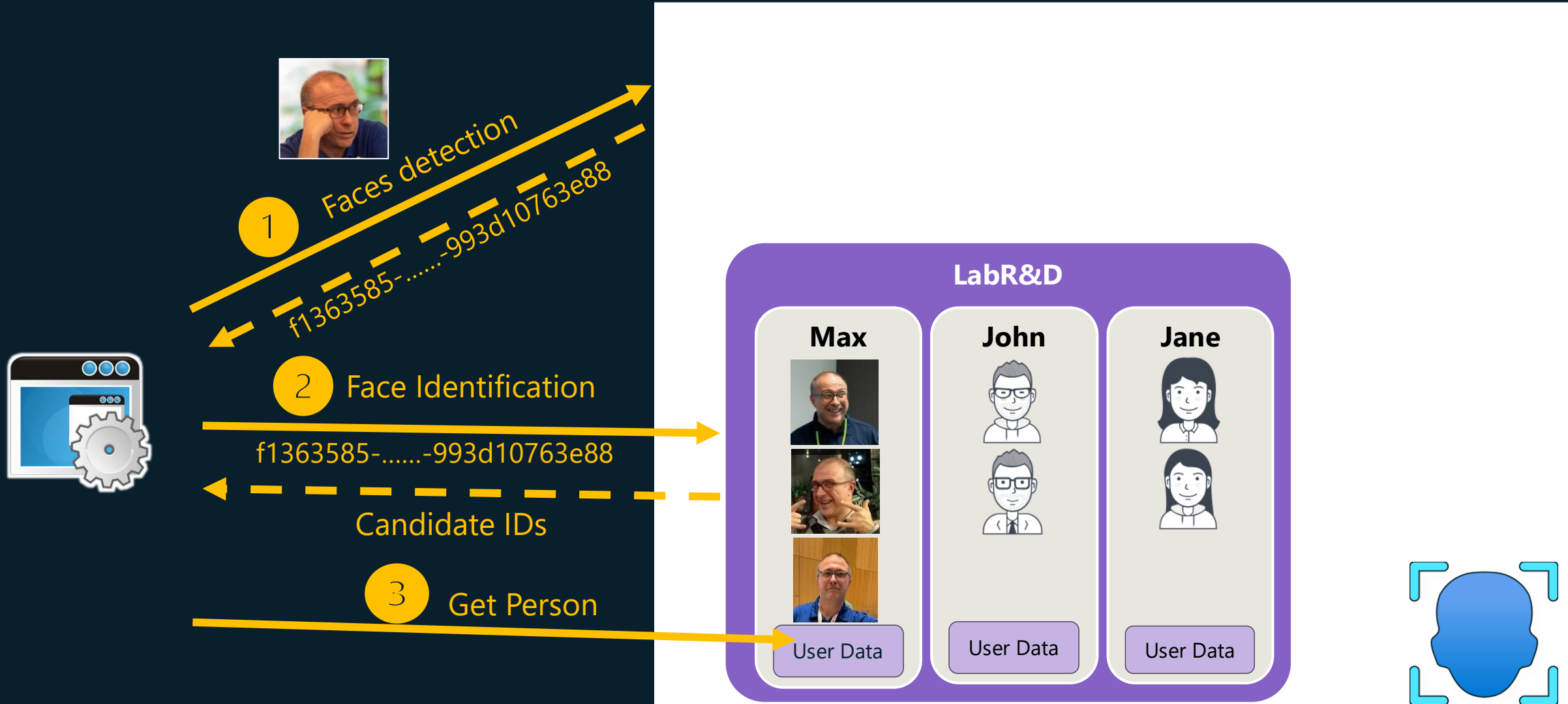
Verification

- Takes a single face ID and a Person object.
- Determines whether the face belongs to that same person.
- Verification is one-to-one matching.

Recognition workflow – define faces knowledge base



Recognition workflow – Identify Persons



Detection and recognition models

Face Service uses two different sets of machine learning models to perform operations on human faces in images.

Detection Models

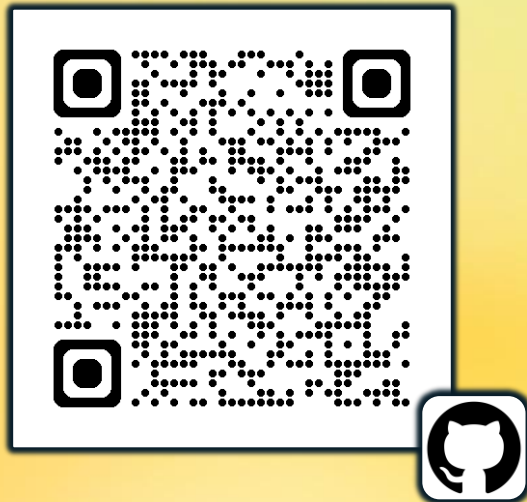
- Detects faces in an image
- 3 models available

Model	Description	Performance notes	Landmarks
detection_01	Default choice for all face detection operations.	Not optimized for small, side-view, or blurry faces.	Returns face landmarks if they're specified in the detect call.
detection_02	Released in May 2019 and available optionally in all face detection operations.	Improved accuracy on small, side-view, and blurry faces.	Doesn't return face landmarks.
detection_03	Released in February 2021 and available optionally in all face detection operations.	Further improved accuracy, including on smaller faces (64x64 pixels) and rotated face orientations.	Returns face landmarks if they're specified in the detect call.

Recognition Models

- Identify, verify, and find similar faces
- Support face enrollment and group training
- 4 models available: from recognition_01 to recognition_04 (faces with mask)

Demo – FACEIT

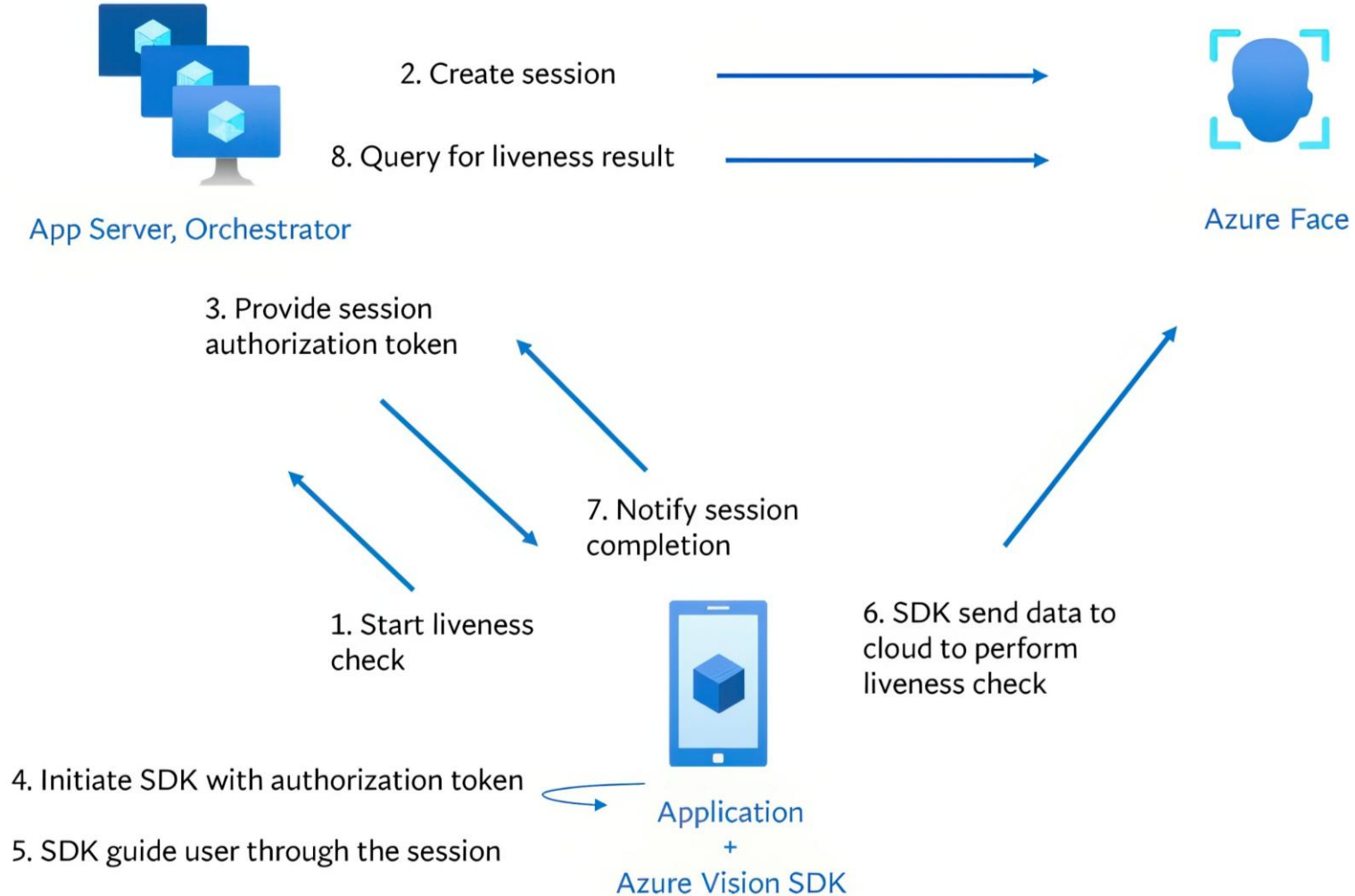


Liveness detection

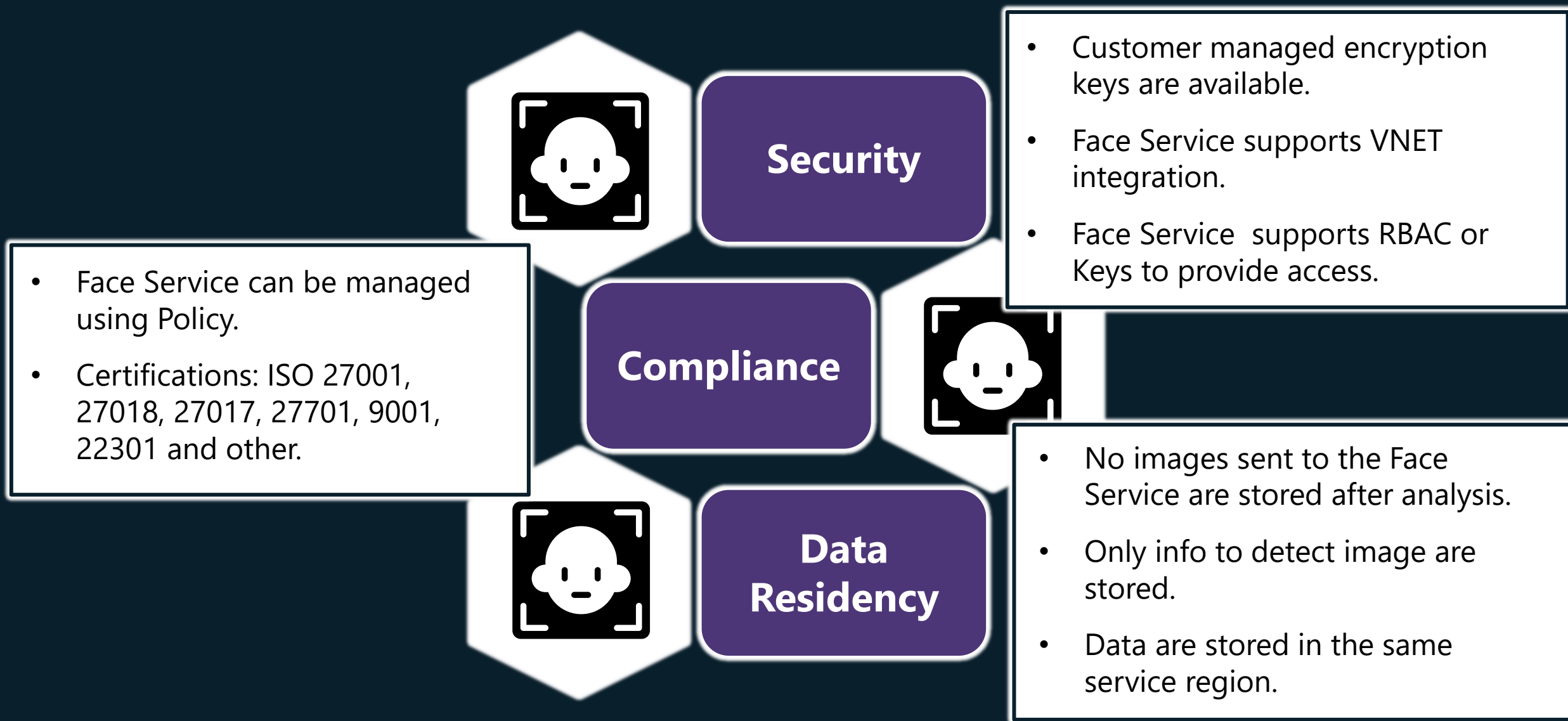
- **Face Liveness** detection can be used to determine if a face in an input video stream is real (live) or fake (spoof).
- The goal of liveness detection is to ensure that the system is interacting with a **physically present live person** at the time of authentication.
- Liveness detection achieved a 0% penetration rate in **iBeta Level 1 and Level 2 Presentation Attack Detection (PAD)** tests



Perform liveness detection



Security, compliance and Data Residency



Pricing



Face Detection	0-1M transactions - €0.859 per 1,000 transactions
Face Verification	
Face Identification	1-5M transactions - €0.687 per 1,000 transactions
Data Structure Management	
Face Grouping	5-100M transactions - €0.516 per 1,000 transactions
Similar Face Search	
Session Management	100M+ transactions - €0.344 per 1,000 transactions
Face Storage	€0.009 per 1,000 faces per month
Train Person Group/Array	€0.859 per 1,000,000 faces
Face Liveness	€12.880 per 1,000 transactions
Face Liveness + Verification	€13.310 per 1,000 transactions

Thank you



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aka.ms/maxlinkedin

Thank you



Data Saturdays Sofia

See You Next Event

References

- [What is the Azure AI Face service? - Azure AI services](#)
- [Are You Alive: Enhancing Azure AI Vision Face API with Liveness Detection](#)
- [Fundamentals of Facial Recognition - Training](#)
- [Understanding Azure Face Service - iEngage 2.0](#)
- [massimobonanni/FACEIT: FACEIT \(Facial Access Control & Environmental Identity Technology\)](#)

